

Lecture 7: Confidence intervals

Economics 326 — Econometrics II

Vadim Marmer, UBC

Point estimation

- Our model:
 1. $Y_i = \beta_0 + \beta_1 X_i + U_i$, $i = 1, \dots, n$.
 2. $E[U_i | \mathbf{X}] = 0$ for all i 's.
 3. $E[U_i^2 | \mathbf{X}] = \sigma^2$ for all i 's.
 4. $E[U_i U_j | \mathbf{X}] = 0$ for all $i \neq j$.
 5. U 's are jointly normally distributed conditional on $\mathbf{X}$.
- The OLS estimator $\hat{\beta}_1$ is a **point estimator** of β_1 .
- With probability **one**, we have that $\hat{\beta}_1 \neq \beta_1$.
- To construct interval estimators, we need to know the distribution of $\hat{\beta}_1$.

Normal distribution

- A normal rv is a continuous rv that can take on any value. The PDF of a normal rv X is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \text{ where}$$

$$\mu = E[X] \text{ and } \sigma^2 = \text{Var}(X).$$

We usually write $X \sim N(\mu, \sigma^2)$.

- If $X \sim N(\mu, \sigma^2)$, then $a + bX \sim N(a + b\mu, b^2\sigma^2)$.

Standard normal distribution

- Standard normal rv has $\mu = 0$ and $\sigma^2 = 1$. Its PDF is $\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right)$.
- Symmetric around zero (mean): if $Z \sim N(0, 1)$, $P(Z > z) = P(Z < -z)$.
- Thin tails: $P(-1.96 \leq Z \leq 1.96) = 0.95$.
- If $X \sim N(\mu, \sigma^2)$, then $(X - \mu)/\sigma \sim N(0, 1)$.

Bivariate normal distribution

- X and Y have a bivariate normal distribution if their joint PDF is given by:

$$f(x, y) = \frac{1}{2\pi\sqrt{(1-\rho^2)\sigma_X^2\sigma_Y^2}} \exp\left[-\frac{Q}{2(1-\rho^2)}\right],$$

$$\text{where } Q = \frac{(x-\mu_X)^2}{\sigma_X^2} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} - 2\rho\frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y},$$

$$\mu_X = E[X], \mu_Y = E[Y], \sigma_X^2 = \text{Var}(X), \sigma_Y^2 = \text{Var}(Y), \text{ and } \rho = \text{Corr}(X, Y).$$

Properties of bivariate normal

If X and Y have a bivariate normal distribution:

- $a + bX + cY \sim N(\mu^*, (\sigma^*)^2)$, where

$$\mu^* = a + b\mu_X + c\mu_Y, \quad (\sigma^*)^2 = b^2\sigma_X^2 + c^2\sigma_Y^2 + 2bc\rho\sigma_X\sigma_Y.$$

- $\text{Cov}(X, Y) = 0 \implies X$ and Y are independent.
- Can be generalized to more than 2 variables (multivariate normal).

Normality of the OLS estimator

- Assume that U_i 's are jointly normally distributed conditional on $\mathbf{X}$ (Assumption 5).
- Then $Y_i = \beta_0 + \beta_1 X_i + U_i$ are also jointly normally distributed conditional on $\mathbf{X}$.
- Since $\hat{\beta}_1 = \sum_{i=1}^n w_i Y_i$, where $w_i = \frac{X_i - \bar{X}}{\sum_{i=1}^n (X_i - \bar{X})^2}$ depend only on $\mathbf{X}$, $\hat{\beta}_1$ is also normally distributed conditional on $\mathbf{X}$.
- Conditionally on $\mathbf{X}$:

$$\hat{\beta}_1 | \mathbf{X} \sim N(\beta_1, \text{Var}(\hat{\beta}_1 | \mathbf{X})),$$
$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

Interval estimation problem

- We want to construct an **interval estimator** for β_1 :
 - The interval estimator is called a **confidence interval** (CI).
 - A CI contains the **true value** β_1 **with some pre-specified probability** $1 - \alpha$, where α is a small probability of error.
 - For example, if $\alpha = 0.05$, then the random CI will contain β_1 with probability 0.95.
- $1 - \alpha$ is called the **coverage probability**.
- Confidence interval: $CI_{1-\alpha} = [LB_{1-\alpha}, UB_{1-\alpha}]$. The lower bound (LB) and upper bound (UB) should depend on the coverage probability $1 - \alpha$.
- The formal definition of CI: It is a **random interval** $CI_{1-\alpha}$ such that conditionally on $\mathbf{X}$,

$$P(\beta_1 \in CI_{1-\alpha} | \mathbf{X}) = 1 - \alpha.$$

Note that the random element is $CI_{1-\alpha}$.

- Sometimes, a CI is defined as $P(\beta_1 \in CI_{1-\alpha}) \geq 1 - \alpha$.

Symmetric CIs

- One approach to constructing CIs is to consider a **symmetric** interval around the estimator $\hat{\beta}_1$:

$$CI_{1-\alpha} = [\hat{\beta}_1 - c_{1-\alpha}, \hat{\beta}_1 + c_{1-\alpha}]$$

- The problem is choosing $c_{1-\alpha}$ such that $P(\beta_1 \in CI_{1-\alpha} | \mathbf{X}) = 1 - \alpha$.
- In choosing $c_{1-\alpha}$ we will be relying on the fact that given our assumptions and conditionally on $\mathbf{X}$:

$$\hat{\beta}_1 | \mathbf{X} \sim N(\beta_1, \text{Var}(\hat{\beta}_1 | \mathbf{X})),$$
$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

- Note that conditionally on $\mathbf{X}$:

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\text{Var}(\hat{\beta}_1 | \mathbf{X})}} \sim N(0, 1).$$

Standard normal quantiles

- Let $Z \sim N(0, 1)$. The τ -th **quantile** (percentile) of the standard normal distribution is z_τ such that $P(Z \leq z_\tau) = \tau$.
- **Median:** $\tau = 0.5$ and $z_{0.5} = 0$. ($P(Z \leq 0) = 0.5$).
- If $\tau = 0.975$ then $z_{0.975} = 1.96$. Due to symmetry, if $\tau = 0.025$ then $z_{0.025} = -1.96$.

σ^2 is known (infeasible CIs)

- **Suppose** (for a moment) that σ^2 is known, and we can compute exactly the variance of $\hat{\beta}_1$:

$$\text{Var}(\hat{\beta}_1 | \mathbf{X}) = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

- Consider the following CI:

$$CI_{1-\alpha} = \left[\hat{\beta}_1 - z_{1-\alpha/2} \sqrt{\text{Var}(\hat{\beta}_1 | \mathbf{X})}, \right. \\ \left. \hat{\beta}_1 + z_{1-\alpha/2} \sqrt{\text{Var}(\hat{\beta}_1 | \mathbf{X})} \right].$$

- For example, if $1 - \alpha = 0.95 \Leftrightarrow \alpha = 0.05 \Leftrightarrow z_{1-\alpha/2} = z_{0.975} = 1.96$, and

$$CI_{0.95} = \left[\hat{\beta}_1 - 1.96 \sqrt{\text{Var}(\hat{\beta}_1 | \mathbf{X})}, \right. \\ \left. \hat{\beta}_1 + 1.96 \sqrt{\text{Var}(\hat{\beta}_1 | \mathbf{X})} \right].$$

Infeasible CI validity (σ^2 known)

- We need to show that

$$P(\beta_1 \in CI_{1-\alpha} | \mathbf{X}) = 1 - \alpha.$$

- Next, let $\sigma_{\hat{\beta}_1} = \sqrt{\text{Var}(\hat{\beta}_1 | \mathbf{X})}$. Then:

$$\begin{aligned} \hat{\beta}_1 - z_{1-\alpha/2} \sigma_{\hat{\beta}_1} &\leq \beta_1 \leq \hat{\beta}_1 + z_{1-\alpha/2} \sigma_{\hat{\beta}_1} \\ \Leftrightarrow -z_{1-\alpha/2} \sigma_{\hat{\beta}_1} &\leq \beta_1 - \hat{\beta}_1 \leq z_{1-\alpha/2} \sigma_{\hat{\beta}_1} \\ \Leftrightarrow -z_{1-\alpha/2} \sigma_{\hat{\beta}_1} &\leq \hat{\beta}_1 - \beta_1 \leq z_{1-\alpha/2} \sigma_{\hat{\beta}_1} \\ \Leftrightarrow -z_{1-\alpha/2} &\leq \frac{\hat{\beta}_1 - \beta_1}{\sigma_{\hat{\beta}_1}} \leq z_{1-\alpha/2} \end{aligned}$$

Infeasible CI validity (σ^2 known)

- We have that

$$\begin{aligned} \beta_1 &\in CI_{1-\alpha} \\ \Leftrightarrow -z_{1-\alpha/2} &\leq \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\text{Var}(\hat{\beta}_1 | \mathbf{X})}} \leq z_{1-\alpha/2}. \end{aligned}$$

- Let $Z = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\text{Var}(\hat{\beta}_1 | \mathbf{X})}} \sim N(0, 1)$ conditionally on $\mathbf{X}$.

$$\begin{aligned} & P(-z_{1-\alpha/2} \leq Z \leq z_{1-\alpha/2} | \mathbf{X}) \\ &= P(z_{\alpha/2} \leq Z \leq z_{1-\alpha/2} | \mathbf{X}) \\ &= 1 - \alpha/2 - \alpha/2 = 1 - \alpha. \end{aligned}$$

Feasible CIs (σ^2 unknown)

- Since σ^2 is unknown, we must estimate it from the data:

$$s^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{U}_i^2 = \frac{1}{n-2} \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2.$$

- We can replace σ^2 by s^2 ; however, the result does not have a normal distribution anymore:

$$\begin{aligned} & \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\widehat{\text{Var}}(\hat{\beta}_1)}} \sim t_{n-2}, \\ & \text{where } \widehat{\text{Var}}(\hat{\beta}_1) = \frac{s^2}{\sum_{i=1}^n (X_i - \bar{X})^2}. \end{aligned}$$

Here t_{n-2} denotes the t -distribution with $n-2$ degrees of freedom.

- The degrees of freedom depend on
 - the sample size (n),
 - and the number of parameters one has to estimate to compute s^2 (two in this case, β_0 and β_1).

Feasible CIs (σ^2 unknown)

- Let $t_{df,\tau}$ be the τ -th quantile of the t -distribution with the number of degrees of freedom df : If $T \sim t_{df}$ then

$$P(T \leq t_{df,\tau}) = \tau.$$

- Similarly to the normal distribution, the t -distribution is centered at zero and is symmetric around zero: $t_{n-2,1-\alpha/2} = -t_{n-2,\alpha/2}$.
- We can now construct a feasible confidence interval with $1 - \alpha$ coverage as:

$$\begin{aligned} CI_{1-\alpha} &= \left[\hat{\beta}_1 - t_{n-2,1-\alpha/2} \sqrt{\widehat{\text{Var}}(\hat{\beta}_1)}, \right. \\ & \quad \left. \hat{\beta}_1 + t_{n-2,1-\alpha/2} \sqrt{\widehat{\text{Var}}(\hat{\beta}_1)} \right], \\ & \text{where } \widehat{\text{Var}}(\hat{\beta}_1) = \frac{s^2}{\sum_{i=1}^n (X_i - \bar{X})^2}. \end{aligned}$$

Example: Rent rates and average income

- Data (RENTAL.DTA): 64 cities in 1990, Rent = average rent, AvgInc = per capita income: $\text{Rent}_i = \beta_0 + \beta_1 \text{AvgInc}_i + U_i$.
- $t_{62,0.975} \approx 2.00 \implies$ The **95%** confidence interval for β_1 is $[0.0115 - 2 \times 0.0013, 0.0115 + 2 \times 0.0013] = [0.0089, 0.0141]$.
- $t_{62,0.95} \approx 1.671 \implies$ The **90%** confidence interval for β_1 is $[0.0115 - 1.671 \times 0.0013, 0.0115 + 1.671 \times 0.0013] = [0.0093, 0.0137]$.

The effect of estimating σ^2

- The t -distribution has heavier tails than the normal.
- $t_{df, 1-\alpha/2} > z_{1-\alpha/2}$, but as df increases $t_{df, 1-\alpha/2} \rightarrow z_{1-\alpha/2}$.
- When the sample size n is large, $t_{n-2, 1-\alpha/2}$ can be replaced with $z_{1-\alpha/2}$.

Interpretation of confidence intervals

- The confidence interval $CI_{1-\alpha}$ is a function of the **sample** $\{(Y_i, X_i) : i = 1, \dots, n\}$, and therefore is **random**. This allows us to talk about the probability of $CI_{1-\alpha}$ containing the true value of β_1 .
- Once the confidence interval is computed given the data, we have its **one realization**. The realization of $CI_{1-\alpha}$ (the computed confidence interval) is not random, and it does not make sense anymore to talk about the probability that it includes the true β_1 .
- **Once the confidence interval is computed, it either contains the true value β_1 or it does not.**